

Software Engineering for Autonomous Systems

Second assignment: student projects

PROJECT DESCRIPTION

Students are expected to design and implement a software system consisting of all the elements defined in the MAPE-K loop shown in Figure 1, i.e., monitoring, analysis, planning, and evolution, using some knowledge. It is also necessary to design the means for specifying high-level goals, which can then be translated into concrete actions performed by actuators.

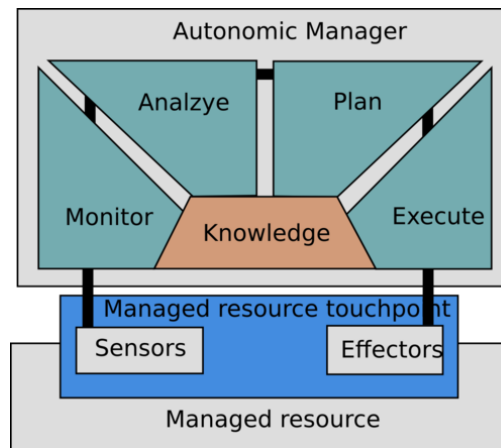


Figure 1: MAPE-K loop

Students are free to identify the application domains and the functionalities of the system to be developed.

The project description shall include the following elements:

- Goals of the system
 - o e.g., the system should manage the environmental quality of a given building with respect to potentially different factors (e.g., power consumption, temperature, light, humidity etc.)
- Describe the managed resources
 - o E.g., heating/cooling systems, smart light bulbs, smart windows, etc.
- Describe the sensors and the effectors for each managed resources
- Discuss the employed architectural pattern for the development of the autonomic manager (see slides 21 and 32-35 of *03-Self-Adaptive-Systems.pdf*)
 - o E.g. single autonomic manager that governs a large part of the system, or several more localised managers making decisions that emerge a global solution to maintain a goal
- Description of the adaptation goals of the autonomic manager
 - o See for example, slide 18 of *Exemplar4.pdf*
 - o Mention the monitored and analyzed characteristics (immediately available or induced by some reasoning system) that are used to support the adaptation goal
- Discuss the approach that has been employed to develop the decision function of the autonomic manager
 - o E.g., search-based, rule-based, AI-based etc.
 - o See *Decision-Function.pdf*

To organise the presentation of the previous point, students can organise the presentation by following the taxonomy on slide 6 of *03-Self-Adaptive-Systems.pdf*

PROJECT DEVELOPMENT AND SUBMISSION PROCESS

1. Student teams are expected to write two pages and send them to davide.diruscio@univaq.it, presenting the autonomous software systems they would like to implement. The documents should clearly describe the proposed systems concerning the elements in Figure 1
2. Once get a positive answer from the docent, teams can start implementing the agreed software systems
3. The teams that want to present and discuss the developed systems should make the source code available to the docent that will perform a first evaluation of the systems offline. The provided software packages should include the required documentation presenting the steps to be performed to have the system running
4. The teams and the docent will agree on a date when the developed systems can be presented and discussed.